404 Mains Electricity

4.4.1 Typical power ratings of appliances

At the textbook's rated 220 V, example powers are 2000 W for a heater, 1200 W for an oven, 1000 W for an air conditioner, 120 W for a refrigerator and 100 W for a television. These illustrate orders of magnitude, not universal constants. Use an actual appliance's rating or the question's supplied data.

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4.4.2 Live, neutral and earth wires

In the ideal mains model used here, neutral is approximately at earth potential while live alternates between positive and negative potentials, providing a voltage between live and neutral. Neutral belongs to the normal operating circuit. The earth wire primarily supplies a protective fault path; their functions are different.

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4.4.3 Household electrical protection

A fuse or circuit breaker interrupts the live supply when current is excessive, reducing overheating; its rating should be just above normal operating current. Earthing a metal case provides a low-resistance fault path if live touches it, helping protection disconnect the supply. Double insulation separates live parts using two insulating barriers. These measures serve different purposes.

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4.4.4 · 4.4.5 · 4.4.6 Alternating current and r.m.s. values

4.4.4 Direct and alternating current

Direct current keeps the same direction although its magnitude may vary. Alternating current reverses periodically, with corresponding voltage polarity changes. Inspect whether current crosses zero and reverses, rather than merely whether it fluctuates. A pulsating current that never reverses is still direct current.

4.4.5 R.m.s. values and alternating-current power

R.m.s. current or voltage is the equivalent d.c. value producing the same average heating power in the same fixed resistance. For a purely resistive load, voltage and current are in phase, so their r.m.s. product gives average power. Zero mean current over a cycle does not mean zero energy dissipation.

$$\bar{P} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} = V_{\text{rms}} I_{\text{rms}}$$

Pure constant resistance R (Ω), so voltage and current are in phase. Average power $\bar{P}$: W; RMS voltage: V; RMS current: A.

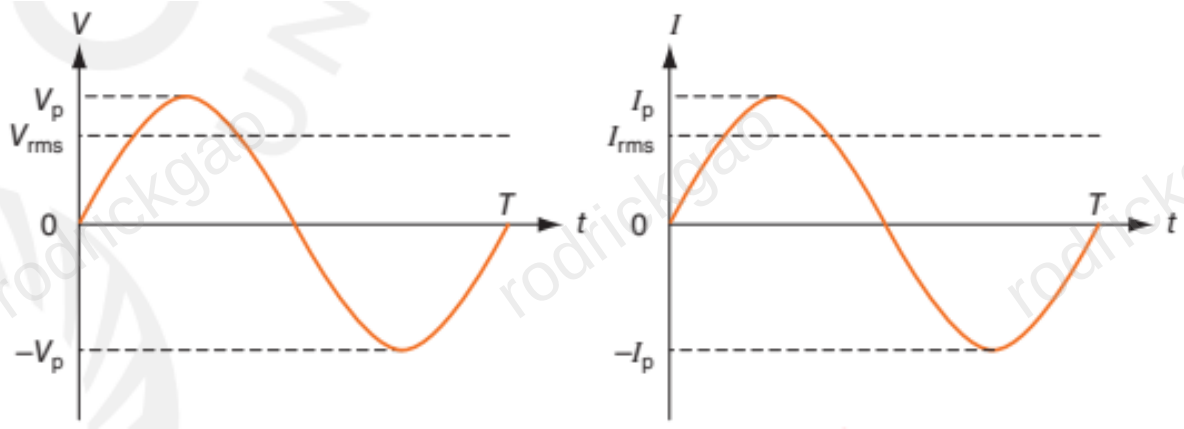
4.4.6 Peak and r.m.s. values of a sinusoid

For a sinusoid with no d.c. offset, the r.m.s. value equals peak value divided by the square root of two. Peak is measured from zero to the maximum magnitude; peak-to-peak is twice the peak value. This conversion is waveform-specific: square waves and other shapes require their own r.m.s. calculation.

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$$

$$I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}}$$

Only for sinusoidal voltage/current with zero DC offset. Peak means amplitude, not peak-to-peak.



Zero lines, peak values and periods of sinusoidal a.c.; r.m.s. value is not peak-to-peak value.

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